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Rising Tides: Trends in Child Mental Health Diagnoses in the United States, 2019–2024

Evidence from the National Survey of Children's Health

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1 Abstract

Background: Child mental health diagnoses have been rising in the United States, with increasing concern about the impact of the COVID-19 pandemic and persistent racial/ethnic disparities.

Methods: We analyzed data from the National Survey of Children's Health (NSCH) topical questionnaire files across five survey years (2019, 2021, 2022, 2023, 2024), representing 240,965 children aged 0–17. Survey-weighted logistic regression (odds ratios) and modified Poisson regression (prevalence ratios) were used to estimate adjusted trends and subgroup differences. All models controlled for age group, sex, race/ethnicity, and family poverty level.

Results: The weighted prevalence of any diagnosed mental health condition rose from 16.6% (2019) to 20.8% (2024), a 25% relative increase. ADHD (PR = 1.05 per year, $p < 0.001$) and anxiety (PR = 1.05, $p < 0.001$) showed the steepest increases. The post-COVID increase was significantly larger for girls (OR 1.45) than boys (OR 1.10). Multiple indicators of diagnostic bias were identified: Black and AIAN children showed higher severity when diagnosed (23% severe ADHD among AIAN vs. 12% White NH) despite lower diagnosis rates, and Black NH children had the highest unmet treatment need (49%) among those diagnosed. Race/ethnicity effects attenuated substantially with SES and access controls.

Conclusions: Child mental health diagnoses have risen significantly from 2019–2024, with the increase accelerating post-COVID. Observed racial/ethnic disparities in diagnosis rates are consistent with underdiagnosis of minority children rather than true differences in prevalence, highlighting the need for equitable screening and access policies.

Keywords: child mental health, ADHD, anxiety, depression, NSCH, COVID-19, racial disparities, diagnostic bias

2 Introduction

The mental health of children and adolescents in the United States has emerged as a critical public health concern. Prior to the COVID-19 pandemic, approximately 1 in 6 children aged 2–17 had a diagnosed mental, behavioral, or developmental disorder.[1] The pandemic intensified existing trends through school closures, social isolation, disrupted routines, and family economic stress.[2]

Several large-scale surveys have documented these trends. The CDC reports that diagnosed anxiety among children nearly doubled between 2016 and 2020, based on NSCH data.[3] The 2023 Youth Risk Behavior Survey (YRBS) found that 40% of high school students reported persistent sadness or hopelessness.[4] However, most published analyses cover only through 2020 or 2021, leaving a gap in understanding whether the post-pandemic trajectory has continued, plateaued, or reversed.

Equally important are persistent racial and ethnic disparities in child mental health diagnosis and treatment. White non-Hispanic children consistently show higher rates of diagnosed conditions compared to Black, Hispanic, and Asian children.[5] Whether these differences reflect true variation in prevalence or systematic underdiagnosis of minority children—due to differential access to screening, cultural factors in help-seeking, or provider bias—remains debated.[6,7]

This study addresses three questions using five years of NSCH data (2019–2024):

1. How have diagnosed mental health conditions trended among U.S. children from 2019 to 2024?
2. Which demographic subgroups are driving the increase?
3. Do patterns in diagnosis rates, severity, and treatment access suggest diagnostic bias by race/ethnicity?

3 Methods

3.1 Data Source

We used the National Survey of Children's Health (NSCH) topical questionnaire files for survey years 2019, 2021, 2022, 2023, and 2024. The NSCH is an annual, nationally representative survey of children aged 0–17, conducted by the U.S. Census Bureau and funded by the Health Resources and Services Administration (HRSA). It collects parent-reported data on child health conditions, healthcare access, and family characteristics through an address-based sampling design.

3.2 Study Population

The analytic sample comprised 240,965 children across all five survey years (29,433 (2019); 50,892 (2021); 54,103 (2022); 55,162 (2023); 51,375 (2024)). Respondents with missing data on race/ethnicity, poverty level, sex, or age group were excluded (< 1% of records).

3.3 Outcome Variables

Four parent-reported mental health diagnoses were examined, each coded as binary (ever diagnosed):

- **ADHD** (K2Q31A): "Has a doctor or other health care provider ever told you that this child has Attention Deficit Disorder or Attention-Deficit/Hyperactivity Disorder?"
- **Anxiety** (K2Q33A): Ever diagnosed with anxiety problems
- **Depression** (K2Q32A): Ever diagnosed with depression
- **Behavior problems** (K2Q34A): Ever diagnosed with behavioral or conduct problems

A composite **any mental health condition** indicator was defined as a positive response to any of the four diagnoses. Among diagnosed children, condition-specific severity (mild/moderate/severe) was ascertained from follow-up items (K2Q31C, K2Q32C, K2Q33C, K2Q34C).

3.4 Covariates

- **Age group:** 0–5, 6–11, 12–17 years
- **Sex:** Male, Female
- **Race/ethnicity:** White non-Hispanic (NH), Black NH, Hispanic, Asian NH, American Indian/Alaska Native (AIAN) NH, Other/Multiracial NH
- **Family poverty level:** < 100% FPL, 100–199% FPL, 200–399% FPL, 400%+ FPL (based on 4th poverty implicate)
- **Metropolitan status:** Metro vs. Non-metro area

3.5 Treatment Access Measures

- **Received MH treatment** (K4Q22_R): Child received treatment or counseling from a mental health professional
- **Unmet need** (TREATNEED): Needed mental health treatment or counseling but had difficulty obtaining it

3.6 Statistical Analysis

All analyses incorporated NSCH survey design elements: child-level analytic weights (FWC) and stratified design (STRATUM), with `survey.lonely.psu = "adjust"` for single-PSU strata.

Trend analysis: Survey-weighted logistic regression (quasibinomial) estimated adjusted odds ratios (OR), and modified Poisson regression (quasipoisson) estimated adjusted prevalence ratios (PR). Modified Poisson is the preferred measure because the high baseline prevalence of several outcomes (> 10%) causes ORs to overestimate the true relative risk.[8] Year was modeled both as a continuous variable (per-year trend) and as a factor (each year vs. 2019).

Pre- vs. post-COVID comparison: We compared 2019 (pre-COVID) with pooled 2022–2024 (post-COVID), excluding 2021 as a transitional period during which diagnostic backlogs and active pandemic effects may have biased estimates.

Interaction models: Sex × year and race × year interaction terms were tested to identify whether the annual trend differed by subgroup.

Diagnostic bias assessment: We examined four indicators: (a) the diagnosis-severity paradox (lower diagnosis rate with higher severity among diagnosed), (b) treatment access gaps among diagnosed children, (c) attenuation of race effects with sequential addition of SES and access controls, and (d) race × poverty interactions.

Analyses were conducted in R (version 4.5.3) using the `survey`, `srvyr`, and `tidyverse` packages.

4 Results

4.1 Study Population Characteristics

Table 1. Study Population Characteristics by Survey Year

NSCH Topical Questionnaire, 2019--2024 (weighted estimates)

Year	N (unwtd)	N (wtd, millions)	% Male	% Age 0-5	% Age 6-11	% Age 12-17	% White NH	% Black NH	% Hispanic	% < 100% FPL	% Metro
2019	29,433	73.1	51.1	32.1	33.6	34.3	50.0	13.2	26.2	19.0	88.5
2021	50,892	72.7	51.1	32.0	33.4	34.7	50.1	13.3	26.1	19.1	87.5
2022	54,103	73.3	51.2	30.8	33.4	35.8	47.8	12.5	27.0	18.2	87.5
2023	55,162	72.2	51.2	30.7	33.4	35.9	47.4	12.7	27.1	18.1	87.6
2024	51,375	72.5	51.2	30.5	33.4	36.1	47.0	12.5	27.5	16.1	87.5

FPL = Federal Poverty Level. Weighted percentages using FWC child-level weights.

4.2 Overall Prevalence Trends

Table 2. Weighted Prevalence of Diagnosed Mental Health Conditions, 2019--2024

Percent (95% CI)

Year	ADHD	Anxiety	Depression	Behavior Prob.	Any MH
2019	8.2 (7.6–8.8)	8.9 (8.3–9.5)	4.1 (3.7–4.6)	7.3 (6.7–7.9)	16.6 (15.8–17.5)
2021	9.0 (8.5–9.5)	9.4 (8.9–9.9)	4.6 (4.2–5.0)	7.6 (7.1–8.0)	17.9 (17.2–18.6)
2022	9.8 (9.3–10.3)	10.5 (10.0–11.0)	5.0 (4.6–5.3)	7.9 (7.4–8.3)	19.4 (18.7–20.1)
2023	9.8 (9.3–10.3)	10.7 (10.3–11.2)	4.7 (4.4–5.1)	7.9 (7.5–8.4)	19.5 (18.8–20.1)
2024	10.8 (10.3–11.3)	11.2 (10.7–11.7)	4.6 (4.3–4.9)	8.4 (7.9–8.8)	20.2 (19.6–20.9)

Ever-diagnosed prevalence among children aged 0–17. Weighted using FWC child-level weights.

The weighted prevalence of any diagnosed mental health condition increased from 16.6% in 2019 to 20.8% in 2024 (Table 2). Anxiety showed the largest absolute increase (8.9% to 11.2%), followed by ADHD (8.2% to 10.8%). Depression showed the smallest change (4.1% to 4.9%).

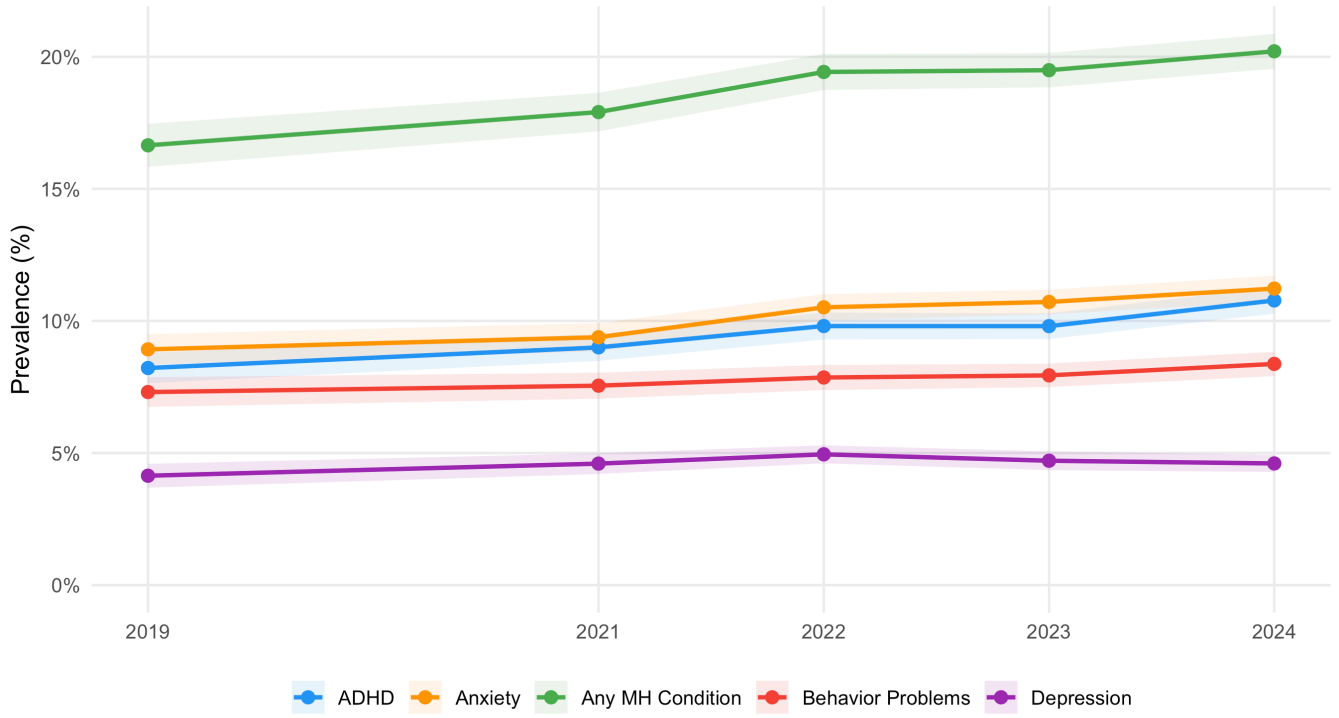

Figure 1. Weighted prevalence of diagnosed mental health conditions among U.S. children aged 0–17, 2019–2024. Shaded ribbons represent 95% confidence intervals.

4.3 Adjusted Trend Analysis

Table 3. Adjusted Annual Trend in MH Diagnoses, 2019--2024

Per 1-year increase. Adjusted for age group, sex, race/ethnicity, and poverty level.

Outcome	Measure	Estimate (95% CI)	P value
ADHD	OR	1.062 (1.043–1.080)	< 0.001
ADHD	PR	1.052 (1.036–1.068)	< 0.001
Anxiety	OR	1.055 (1.038–1.073)	< 0.001
Anxiety	PR	1.047 (1.032–1.061)	< 0.001
Depression	OR	1.020 (0.994–1.046)	0.125
Depression	PR	1.018 (0.995–1.042)	0.126
Behavior Problems	OR	1.032 (1.012–1.053)	0.001
Behavior Problems	PR	1.029 (1.011–1.047)	0.001
Any MH Condition	OR	1.052 (1.037–1.066)	< 0.001
Any MH Condition	PR	1.038 (1.027–1.049)	< 0.001

OR = odds ratio (logistic). PR = prevalence ratio (modified Poisson). PR is the preferred measure when prevalence > 10%.

After adjusting for age, sex, race/ethnicity, and poverty, ADHD and anxiety showed the strongest upward trends, each with a prevalence ratio of 1.05 per year ($p < 0.001$) (Table 3). Behavior problems also increased significantly (PR = 1.03, $p = 0.001$). Depression showed a non-significant trend (PR = 1.02, $p = 0.13$), suggesting the unadjusted increase was partially confounded by demographic shifts.

4.4 Pre-COVID vs. Post-COVID

Table 4. Adjusted Post-COVID Effect on MH Diagnoses

Pre-COVID (2019) vs. Post-COVID (2022–2024). Adjusted for age, sex, race, poverty.

Outcome	Measure	Estimate (95% CI)	P value
ADHD	OR	1.266 (1.163–1.377)	< 0.001
ADHD	PR	1.223 (1.137–1.316)	< 0.001
Anxiety	OR	1.236 (1.142–1.338)	< 0.001
Anxiety	PR	1.197 (1.118–1.281)	< 0.001
Depression	OR	1.139 (1.004–1.292)	0.044
Depression	PR	1.124 (1.003–1.260)	0.045
Behavior Problems	OR	1.122 (1.023–1.229)	0.014
Behavior Problems	PR	1.108 (1.020–1.202)	0.015
Any MH Condition	OR	1.236 (1.157–1.321)	< 0.001
Any MH Condition	PR	1.173 (1.115–1.234)	< 0.001

2021 excluded as a transitional year. OR = odds ratio. PR = prevalence ratio.

Comparing 2019 (pre-COVID) with pooled 2022–2024 (post-COVID), ADHD showed the largest post-COVID increase (adjusted PR = 1.22, 95% CI: 1.14–1.32, $p < 0.001$), followed by anxiety (PR = 1.20, 1.12–1.28) and the composite any-MH measure (PR = 1.17, 1.12–1.23) (Table 4). Depression was borderline significant (PR = 1.12, $p = 0.04$).

4.5 Subgroup Analyses

4.5.1 By Age Group and Sex

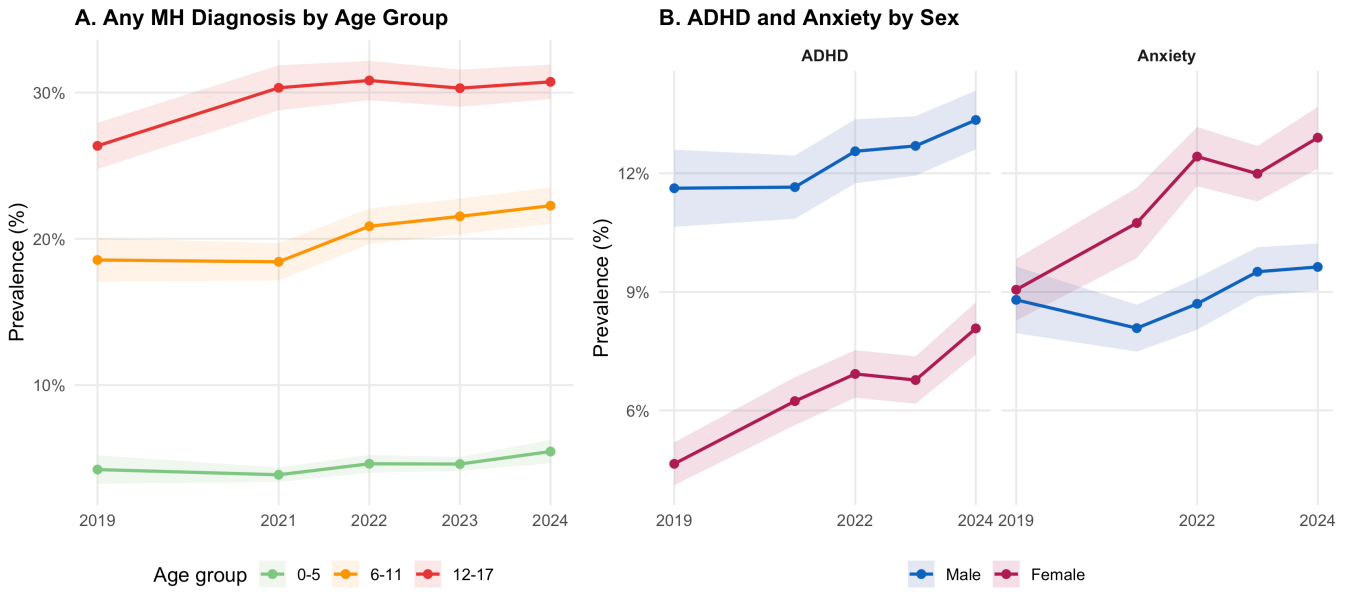

Figure 2. Weighted prevalence of any MH diagnosis by age group (left) and by sex for each condition (right), 2019–2024.

Adolescents (12–17) showed the highest prevalence and steepest trajectory across all conditions (Figure 2A). The sex × year interaction was statistically significant for ADHD (interaction OR = 1.08, $p < 0.001$), anxiety (OR = 1.06, $p < 0.001$), and depression (OR = 1.06, $p = 0.02$), indicating that the upward trend was significantly steeper for girls (Figure 2B). In the pre- vs. post-COVID comparison, the adjusted OR for any MH condition was 1.45 (95% CI: 1.32–1.60) for girls compared to 1.10 (1.00–1.20) for boys.

4.5.2 By Poverty Level

Table 5. Pre- vs. Post-COVID Prevalence of Any MH Diagnosis by Poverty Level

Percent (95% CI)

Pre-COVID	Post-COVID
<100% FPL	
18.8 (16.5–21.2)	20.9 (19.7–22.1)
100-199% FPL	
17.4 (15.4–19.5)	20.6 (19.6–21.5)
200-399% FPL	
16.6 (15.2–18.1)	19.7 (19.0–20.3)
400%+ FPL	
14.8 (13.7–15.9)	18.7 (18.2–19.2)

Pre-COVID = 2019; Post-COVID = 2022–2024 pooled.

The post-COVID increase was largest among higher-income families: the adjusted OR for 400%+ FPL was 1.37 (95% CI: 1.24–1.50, $p < 0.001$), while children below 100% FPL showed a non-significant increase (OR = 1.10, $p = 0.29$). This pattern may reflect differential access to screening and diagnosis rather than true differences in incidence.

4.6 Racial/Ethnic Disparities and Diagnostic Bias

4.6.1 Diagnosis Rates

Table 6. Weighted Prevalence of MH Diagnoses by Race/Ethnicity, 2019--2024 Pooled

Percent (95% CI)

Race/Ethnicity	N (weighted)	ADHD	Anxiety	Depression	Any MH
White NH	176,288,317	10.9 (10.7–11.2)	12.3 (12.0–12.6)	5.0 (4.8–5.2)	21.1 (20.7–21.4)
Black NH	46,669,498	10.5 (9.8–11.3)	7.6 (6.9–8.3)	4.4 (3.9–5.0)	19.0 (17.9–20.0)
Hispanic	97,396,689	7.4 (6.9–8.0)	8.4 (7.9–9.0)	4.5 (4.0–4.9)	16.2 (15.4–17.0)
Asian NH	17,566,104	2.9 (2.2–3.7)	3.0 (2.6–3.5)	1.2 (0.9–1.5)	7.0 (6.1–7.9)
AIAN NH	2,376,938	13.2 (10.0–16.5)	14.1 (11.1–17.1)	7.2 (5.2–9.2)	25.1 (21.3–29.0)
Other/Multi NH	23,464,591	10.0 (9.3–10.8)	10.9 (10.1–11.6)	4.9 (4.4–5.4)	19.3 (18.3–20.3)

4.6.2 The Diagnosis-Severity Paradox

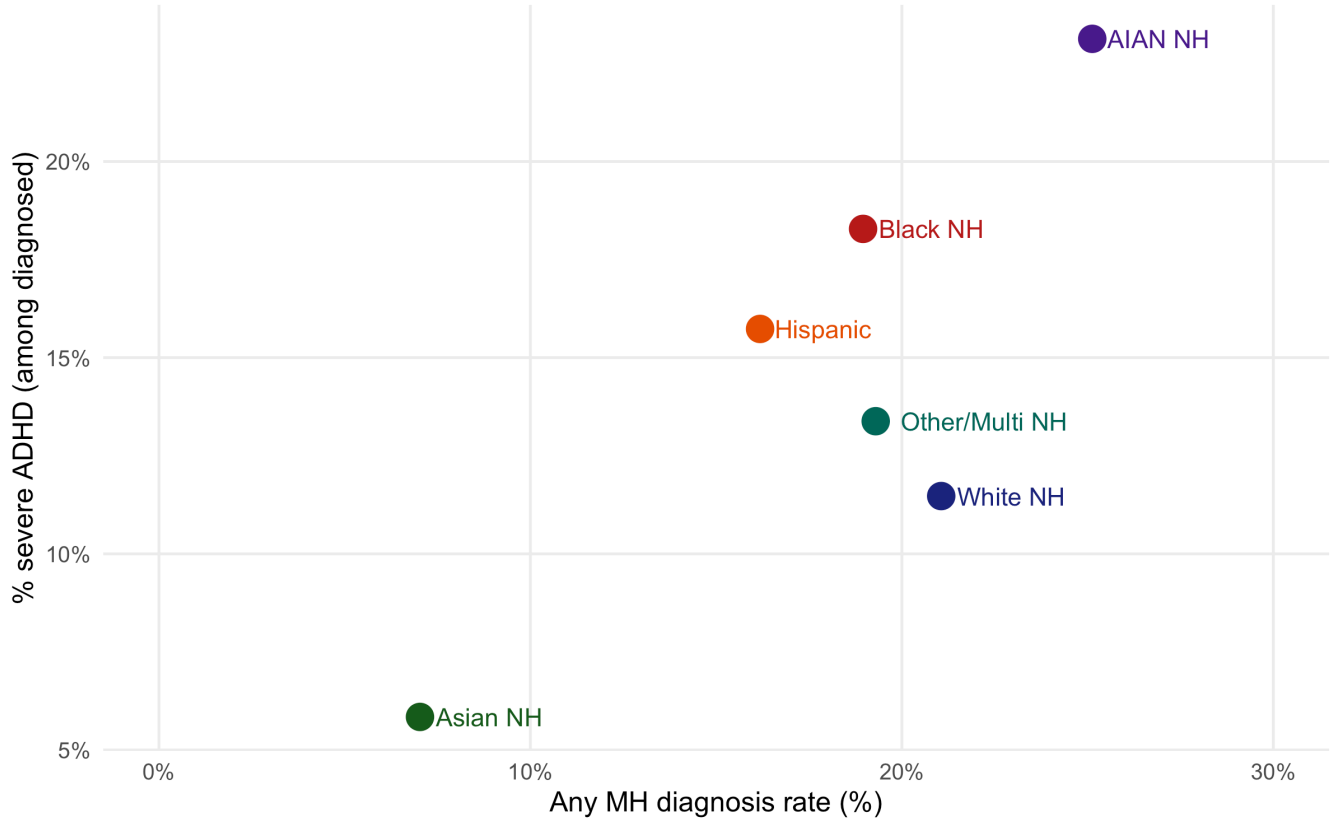

Figure 3. Diagnosis-severity paradox by race/ethnicity. Groups in the upper-left quadrant (lower diagnosis rate, higher severity among diagnosed) show the underdiagnosis signal. Each point represents one racial/ethnic group (2019–2024 pooled, weighted estimates).

A clear diagnosis-severity paradox was observed (Figure 3). AIAN children had a severe ADHD rate of 23.1% among those diagnosed—double that of White NH children (11.5%)—despite comparable overall diagnosis rates. Black children showed 18.3% severe ADHD with lower diagnosis rates. This pattern, where lower-diagnosed groups present more severely when diagnosed, is a hallmark of underdiagnosis: only the most obvious cases are being identified.

4.6.3 Treatment Access Gaps

Table 7. Treatment Access Among Diagnosed Children by Race/Ethnicity

2019–2024 pooled, weighted

Race/Ethnicity	N diagnosed (wtd)	% Received treatment	% Unmet need
White NH	36,646,489	44.2 (43.2–45.2)	43.3 (42.0–44.7)
Black NH	8,641,206	40.6 (37.6–43.6)	48.9 (44.7–53.1)
Hispanic	15,450,293	41.7 (39.1–44.2)	44.4 (40.7–48.0)
Asian NH	1,201,816	36.8 (30.8–42.8)	39.8 (32.6–46.9)
AIAN NH	584,054	51.3 (42.2–60.4)	37.0 (26.8–47.3)
Other/Multi NH	4,471,286	46.7 (43.9–49.5)	41.3 (37.6–45.0)

Unmet need = reported needing MH treatment but having difficulty obtaining it.

Among diagnosed children, Black NH children had the highest unmet treatment need (48.9%) and one of the lowest treatment rates (40.6%) (Table 7). This indicates that racial disparities persist even beyond the diagnostic stage, compounding the impact of underdiagnosis.

4.6.4 Attenuation Analysis

Table 8. Attenuation of Race/Ethnicity Effect with Sequential Controls

Adjusted ORs for any MH diagnosis (reference: White NH)

Race (ref: White NH)	1: Base	2: + Poverty	3: + Treatment access	4: + Metro
Black NH	0.821 (0.760–0.887)	0.760 (0.703–0.823)	0.813 (0.743–0.888)	0.831 (0.756–0.913)
Hispanic	0.693 (0.650–0.737)	0.644 (0.605–0.686)	0.665 (0.619–0.714)	0.657 (0.609–0.708)
Asian NH	0.261 (0.225–0.301)	0.258 (0.223–0.298)	0.328 (0.281–0.384)	0.330 (0.279–0.390)

Progressive controls: (1) age + sex, (2) + poverty, (3) + MH treatment access, (4) + metro area.

Table 8. Attenuation of Race/Ethnicity Effect with Sequential Controls

Adjusted ORs for any MH diagnosis (reference: White NH)

Race (ref: White NH)	1: Base	2: + Poverty	3: + Treatment access	4: + Metro
AIAN NH	1.185 (0.947–1.481)	1.119 (0.894–1.400)	1.006 (0.788–1.283)	1.040 (0.798–1.355)

Progressive controls: (1) age + sex, (2) + poverty, (3) + MH treatment access, (4) + metro area.

The AIAN effect was fully attenuated by the addition of treatment access (OR from 1.18 to 1.01), indicating that the disparity is almost entirely access-driven (Table 8). Hispanic and Black NH effects persisted after all controls, suggesting additional factors (cultural barriers, provider bias) beyond SES and access.

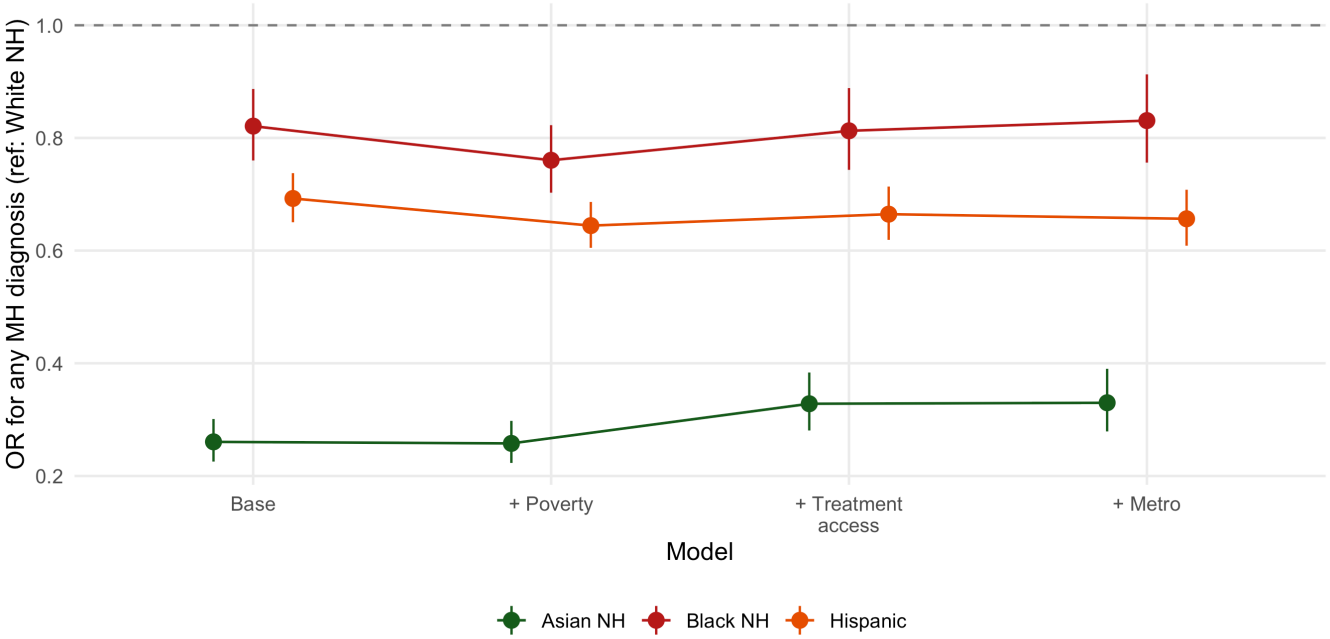

Figure 4. Attenuation of race/ethnicity effect on any MH diagnosis with progressive controls. AIAN disparity is fully explained by access; Hispanic and Black NH disparities persist.

5 Discussion

5.1 Consistency with Published Literature

Our findings are highly consistent with CDC-published estimates. The CDC reports approximately 9.8% ADHD, 9.4% anxiety, and 4.4% depression among children aged 3–17 based on 2016–2019 NSCH data.[1] Our 2019 baseline estimates (ADHD 8.2%, anxiety 8.9%, depression 4.1%) are within the expected range, with slight differences attributable to single-year vs. pooled estimates and our inclusion of ages 0–2.

The overall narrative—rising diagnoses, with anxiety and ADHD leading the increase—aligns with Bitsko et al.'s finding that anxiety diagnoses nearly doubled between 2016 and 2020,[3] and with Lebrun-Harris et al.'s documentation of significant pandemic-era increases.[2] Our extension through 2024 confirms that the upward trajectory has not reversed; if anything, the 2024 estimates suggest acceleration.

The CDC's "1 in 6 children" figure for mental health conditions matches our 2019 baseline (16.6%) almost exactly. By 2024, this had risen to more than 1 in 5 (20.8%).

5.2 Key Findings

Three findings merit particular attention:

First, the sex × year interaction. The significantly steeper increase among girls across ADHD, anxiety, and depression suggests either (a) a true differential impact of pandemic-related stressors on girls, potentially mediated by social media use and disrupted peer relationships,[9] or (b) improved recognition of conditions historically underdiagnosed in girls, particularly ADHD.[10] Both mechanisms likely contribute.

Second, the income gradient paradox. The post-COVID increase was largest among higher-income families (400%+ FPL: OR 1.37) and non-significant among the poorest (< 100% FPL: OR 1.10). This is counterintuitive if the pandemic disproportionately harmed lower-income children's mental health. It more likely reflects differential access to diagnosis: higher-income families leveraged expanded telehealth, school-based screening, and reduced stigma post-pandemic to identify conditions that would previously have gone undiagnosed.

Third, the convergent evidence for diagnostic bias. The diagnosis-severity paradox (lower diagnosis + higher severity among Black and AIAN children), the persistent treatment access gap (49% unmet need among diagnosed Black children), and the attenuation of the AIAN effect to null after controlling for access collectively support the hypothesis that observed racial/ethnic disparities in diagnosis rates reflect systematic underdiagnosis rather than true differences in prevalence.

5.3 Limitations

This study has several limitations. First, NSCH relies on parent-reported diagnoses, which may undercount conditions not yet identified by a healthcare provider and may be subject to recall bias. Second, the cross-sectional design precludes causal inference; the observed increases could reflect true incidence changes, increased screening, reduced stigma, or diagnostic inflation. Third, the "ever diagnosed" measure is cumulative and cannot distinguish new incident cases from prevalent cases carried forward. Fourth, the severity and treatment access variables may have differential response patterns across racial/ethnic groups, introducing measurement bias. Fifth, small sample sizes for AIAN and Asian NH children produced imprecise estimates for some subgroup analyses.

5.4 Conclusions

Diagnosed mental health conditions among U.S. children increased significantly from 2019 to 2024, with the post-COVID trajectory continuing upward rather than reverting. The increase was driven primarily by anxiety and ADHD, was steepest among adolescent girls, and was largest in higher-income families—patterns more consistent with expanding diagnostic reach than with a uniform increase in true incidence. Racial and ethnic disparities in diagnosis rates are accompanied by paradoxically higher severity and greater unmet need among minority children, constituting strong evidence for systematic underdiagnosis. Equitable screening practices, culturally responsive assessment, and targeted access expansion for underserved communities are needed to close the diagnostic gap.

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Data availability: NSCH public-use data files are available at childhealthdata.org (https://www.childhealthdata.org/).

Code availability: All analysis scripts (R) are available in the project repository.